CHAPTER ONE
INTRODUCTION

1.1 Background to the Study

A chatbot is an Artificial Intelligence-based computer program that can replicate the conversion with a user in a human language like a human having a conversation with another human, through messaging applications, websites, mobile applications, or voice over the telephone(Nagarajan, (2019)). As the name chatbot describes, it is a combination of "chat" means conversion or chatting, and "robot" means to replicate human behavior, so the name describes what the computer program does(Adamopoulou, Moussiades, (2020, June)).

Nowadays, chatbots have gained popularity due to their permanent availability as Software-as-a-Service, their improved linguistic features and the improvement of intelligence thanks to machine learning and AI. As a result, chatbots have contributed to one third to half of all online conversations from 2007 to 2015(Tsvetkova et al, 2016; Radziwill, 2017) .Interesting enough, chatbots were actually first designed to find out whether users could be misled into thinking they were human beings(Abu Shawar & Atwell, 2007).

Telecommunications is now widely understood to refer to the electrical means of communicating over long distances(Edan, Idowu,Zango, (2013)). The provision of online customer services by the telecommunication company is effective for solving issues customers encounter during the use of the services provided by service providers. Online customer services are the most effective method when customers cannot be at the office or stations of their service provider(Oduro, Boachie-Mensah, Agyapong, (2018)). For more convenience, emailing and chatting with the customer service representative are the best options for online supported customer services. Chatbots and auto-response replies can be integrated into the customer service rendering system, to improve the latency of response for a need of assistance by the customer. Chatbots are game-changing (Tawk, 2021) because an operator doesn't need to be on the system to give an appropriate response to the customers without delay or the customer being in a queue.It allows humans to interact with technology using a variety of input methods such as voice, text, gesture, and touch, 24 hours a day, seven days a week.

To increase business success, the chatbot does not have to be as perfect as humans or does not have to mimic human behavior (Gadiyar, 2017). The chatbot should only be able to give fast and clear responses to the users. They should fulfill or answer the questions asked by users to some extent.
For many years, chatbots were typically used in customer service environments, but they are now being used in a variety of other roles within enterprises to improve customer experience and business efficiencies(Dahiya, (2017)).

Dmitrii Dumik, the co-founder of Chatfuel, believes that chatbots will turn into a more engaging experience and finally become just as or even more popular than the mobile application is today, providing the user with a more convenient experience(Russell, 2016).
One may not consider chatbots as being one of the newest and most emerging trends in technology and Social Media, however, Schlicht (2016) argues that the four biggest messaging applications have already overtaken the four biggest social networking apps in number of active users, which demonstrates the increasing significance of engaging with customers via messaging applications. Furthermore, the fact that chatbots are faster than most other applications or websites for users to reach their goals illustrate the further importance.
A Chatbot key task is to help users,by providing answers to users, in this case, customers.

1.2 Statement of the Problem

Many customers face difficulties when trying to troubleshoot their network and services provided by the network. A large number of customers who require assistance to reach out to the available customer service representatives makes the customer service rendering congested with so many customers not being attended to or not well assisted with the problem they might face, and this leads to low customer satisfaction rating for business operators (i.e the telecommunication company). MTN Group, in this instance, is a multinational telecommunication company with a wide range of customers in the world.MTN is a popular choice and one of the most used network service providers (Wibowo, (2020)). The number of MTN customer representatives are not able to serve the millions of customers' needs and assistance, and as an inclusion, the chatting platform which it also provides is not well functional in interaction and user experience.

1.3 Aim and Objectives

This project is aimed to design and implement an intelligent and conversational Chatbot system as support to customer service provision and the customer service representatives for MTN Nigeria telecommunication company with the following objectives:

1. To develop the model using NLP(Natural Language Processing),Word tokenizers for an Intelligent Chatbot System.
2. To help in catering for a huge amount of customers at the same time all through the day.
3. To assist customers with little or no computer knowledge and increase customer satisfaction rating**.**

1.4 Significance of the Study

This system will be extremely useful in a variety of cases, including reducing congestions to the available customer service options; and assisting telecommunication companies, which can be useful in a variety of settings, when a user has a minor complaint, user having not too technical assistance, and the shortage of real people customer service representative and as well as in the use of research and developing a chatbot system.

1.5 Scope of the Study

The study is set to cover communication between a chatbot in rendering technical and non-technical assistance to customers. The data used for this study is obtained from real conversations between the MTN customer representative and the customer. The data is used to model a chatbot system that will be able to give solutions in an interactive way to customers.

1.6 Limitations of the Study

This was limited by the resources from data, and time of training the chatbot. Data was sourced from a real conversation with a customer care representative, which makes the data in a few thousands. This system or model will be trained to have a conversational dialogue with customers of a wide range of languages and locations. Hence the research work is limited as well as time was another important factor that limited the extent to which the work wants to research.

1.7 Definition of Terms

Natural language Processing: NLP (as it's also known) is a field of computer science that focuses on creating computers and software that understands human speech and language.

Tokenization: Tokenization is the process by which a large quantity of text is divided into smaller parts called tokens. These tokens are very useful for finding patterns and are considered as a base step for stemming and lemmatization. Tokenization also helps to substitute sensitive data elements with non-sensitive data elements.

Data mining: The data mining process involves a number of steps from data collection to visualization to extract valuable information from large data sets. As mentioned above, data mining techniques are used to generate descriptions and predictions about a target data set.

Multinomial Naive Bayes: The Multinomial Naive Bayes algorithm is a Bayesian learning approach popular in Natural Language Processing (NLP). The program guesses the tag of a text, such as an email or a newspaper story, using the Bayes theorem. It calculates each tag's likelihood for a given sample and outputs the tag with the greatest chance. The Naive Bayes classifier is made up of a number of algorithms that all have one thing in common: each feature being classed is unrelated to any other feature. A feature's existence or absence has no bearing on the inclusion or exclusion of another feature.

Recurrent Neural Networks: A recurrent neural network is a type of artificial neural network commonly used in speech recognition and natural language processing. Recurrent neural networks recognize data's sequential characteristics and use patterns to predict the next likely scenario.

Artificial Intelligence: is best described as the development of a computer system's ability to perform tasks normally, requiring human intelligence, such as visual perception, speech recognition, decision-making and translation between languages.

Algorithms: Are processes or sets of rules to be followed in calculations or other problem-solving operations, especially by a computer.

AIML: Is an XML based programming language which was initially designed to help the development of A.L.I.C.E., created by Richard Wallace, and is a text file with a specific structure which makes up the knowledge base of a chatbot. Nowadays, it is widely used in the development of software agents that communicate with their users using Natural Language. (Burguillo et al., 2009)

CHAPTER TWO

LITERATURE REVIEW

2.1 Theoretical Background

Introduction To The General Subject Area

The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it." (Weiser, 1991)

What can be drawn from this quote is that any technology that is implemented, accepted and used frequently by the human species gets to be taken for granted, while turning into an essential and becoming labelled as a necessity of everyday life. At one point, there will be no point of return, guaranteeing that the mentioned technology will stay part of the lives of everyone. Benton and Radziwill (2017) suggest that this could happen with chatbots.

Benton and Radziwill (2017) describe a chatbot as the impression of interacting with humans online, while actually communicating with a computer software, put to life by natural language input. Others simply define it as a computer program which imitates conversations with users, applying artificial intelligence. (Wong, 2016)

Scharl (2004) explains that a chatbot is a software which typically permits textual communication using natural language. It seems crucial for the user's acceptance of chatbots to simulate real humans, which in turn further highlights the critical importance for a large knowledge base, i.e. the existing set of rules a chatbot has. (Scharl, 2004)

With more and more advertising making it to and overwhelming customers both online and in real life, the difficulty of reaching them turns more and more into a difficult task. Chatbots could soon become one of the best ways for organisations to get in touch with the individual customer and will decide upon a company's competitiveness. (Moore, 2017) Furthermore, key developments concerning the growth of messaging services and the advances in Artificial Intelligence have largely attributed to the recent interest in chatbots (Guzman et al, (2016)).

Historical perspective Of Chatbots

Alan Truing proposed the research paper on the Turing test, He is the creator of the Turing test.Alan Turing posed the question "Can machines think?", the paper was published in 1950 (Turing, 1950). We say the machine can think if the human and machine are indistinguishable.

Technology in conversational bot has advanced so much in the past decades (Wallace, (2009)). The first chatbot was so simple that it only used to give output in text and later in the form of voice.

Weizenbaum (1966), ELIZA was created by Joseph Weizenbaum at MIT,it was arguably the first chatbot that came close to imitating a human. ELIZA would identify keywords in an input sentence and pattern match those keywords against a set of pre-programmed rules to generate appropriate responses. There has been progressing in the development of increasingly intelligent chatbots since ELIZA. ELIZA'S successor is PARRY,Kenneth Colby at Stanford created PARRY in

1972(Pereira et al, (2016)), a bot that impersonated a paranoid schizophrenic.PARRY was in the rare list of chatbots that would pass the Turing test (Bhakta, Savin-Baden, Burden, (2016)). PARRY is similar to ELIZA because it was a fork from it. It was simply a rule-based bot. It has some improvement from ELIZA. PARRY has a better understanding of the human language so it offers a more controlling structure. The advanced feature of PARRY is, it reacts by the understanding of users' tone and it simulates the response.

In 1995, Richard Wallace introduced ALICE (artificial linguistic internet computer entity). ALICE was developed in Java. It was interested in pattern matching to simplify conversations. ALICE used the AIML (Artificial Intelligence Markup Language), which made this bot unique and special. AIML allows creating tags that help ALICE for recursive pattern matching, it also simplifies the input of the language.er, 2004). ALICE could not hold the conversion for a long time, so it does not pass the Turing test.

Amazon's Echo and Alexa, Apple's Siri, and Microsoft's Cortana are examples of modern chatbots. These bots' architectures and retrieval processes leverage advances in machine learning to provide advanced "information retrieval" processes in which responses are generated based on analysis of web search results. Others have responded with "generative" models, which use statistical machine translation (SMT) techniques to "translate" input phrases into output responses. A current best practice is Seq2Seq, an SMT algorithm that uses recurrent neural networks (RNNs) to encode and decode inputs into responses.

Definition Of The Specific Research Concept

There are a variety of approaches to evaluating chatbot performance. Chatbots have specific functions in terms of information retrieval (IR): there are virtual assistants, question-answer bots, and domain-specific bots. Evaluators should ask the chatbot questions and make requests, assessing effectiveness by measuring accuracy, precision, recall, and F-score with the correct chatbot response.

The bot's goal, from a user experience standpoint, is arguably to maximize user satisfaction. Users should be polled (typically via questionnaires on platforms such as Amazon Mechanical Turk), and bots should be ranked based on usability and satisfaction. Bots should approximate speech and be evaluated by linguistic experts on their ability to generate complete, grammatical, and meaningful sentences.

The following concepts are attached to the study and development of chatbots:

Pattern Matching

Pattern Matching is based on blocks of representative stimulus-response data. A sentence (stimuli) is entered, and output (response) is generated following the user input. Eliza and ALICE were the first chatbots created with the help of pattern recognition algorithms. The disadvantage of this approach is that the responses are completely predictable, repetitive, and devoid of human touch. There is also no storage of previous responses, which can lead to looping conversations.

Natural Language Processing

The goal of natural language processing (NLP) is to take the ASR's unstructured output and generate a structured representation of the text that includes spoken language understanding (SLU) or, in the case of text input, natural language understanding (NLU) (NLU). In this section, we look at a variety of methods for extracting semantic information and meaning from the spoken and written language to create grammatical data structures that can be processed by the Dialogue

Management unit in the following step. This is difficult because speech can contain identity-specific encodings in addition to meaning-encodings and (ii) environmental noise. Similarly, both speech and text inputs to a chatbot may include (iii) grammatical errors, (iv) disfluencies, (v) interruptions, and (vi) self-corrections.

Artificial Intelligence Markup Language (AIML)

The Artificial Intelligence Markup Language (AIML) was developed between 1995 and 2000 and is based on Pattern Recognition or Pattern Matching techniques. It is used in natural language modeling for conversation between humans and chatbots using the stimulus-response approach. It is a tag-based, XML-based markup language. AIML is based on basic dialogue units called categories (tag category>), which are formed by user input patterns (tag pattern>) and chatbot responses (tag template>).

Word Embedding

In a nutshell, word embedding is simply the act of mapping words into vectors. This vector representation can then be directly fed into a machine learning algorithm as features. There are multiple ways of performing this operation, ranging from a simple count vector to deep learning approaches such as Word2vec and GloVe. The latter will be of particular interest for this thesis as they have proven to work efficiently in the field of conversational agents. In particular, the skip-gram model will be explained extensively since it is the technique used in the Keras library for embedding layers.

Discussion On the Study Area

Chatbots also known as "conversational agents" are software applications that mimic written or spoken human speech for the purposes of simulating a conversation or interaction with a real person.There are two primary ways chatbots are offered to visitors: via web-based applications or standalone apps. Today, chatbots are used most commonly in the customer service space, assuming roles traditionally performed by living, breathing human beings such as Tier-1 support operatives and customer satisfaction reps(Sarah, (2018)). Chatbots have become extraordinarily popular in recent years largely due to dramatic advancements in machine learning and other underlying technologies such as natural language processing. Today's chatbots are smarter, more responsive, and more useful and we're likely to see even more of them in the coming years(John, (2020)).

2.2 Review Of Relevant Approaches

Machine Learning Approach For Chatbots

Machine learning is defined as the field of study that provides computers with the ability to learn from input data without being explicitly programmed to do so. The learning process is done iteratively from analyzed data and new input data. This iterative aspect allows computers to identify hidden insights and repeated patterns and use these findings to adapt when exposed to a new data (Ali et al., 2019). This approach comprises of three main techniques which are: Supervised learning, Unsupervised learning, Semisupervised.

The Supervised learning includes training datasets at which data is presented in pairs, an input and its corresponding correct output. The training of the model is based on a set data called the training set after which it will be tested on a set of data called testing set.

Some algorithms that can be used in Supervised learning includes: Linear regression, Naïve Bayes, Support Vector Machine and so on (Swamynathan, 2017).

Unsupervised learning is different from the supervised learning, it uses an input dataset without any labeled outputs to train the learning algorithm. There is no correct or incorrect output to each input object and it

does not require any human intervention to correct or adjust as in supervised learning. Thus, Unsupervised learning is more subjective than Supervised Learning. The main goal of unsupervised learning is to learn more about the data through identifying the fundamental structure or distribution patterns that is found in the data itself. Examples of the Unsupervised learning algorithm are K-means which is suitable for Clustering problem and Apriori Algorithm which is as well suitable for association problems (Swamynathan, 2017).

The Semi-supervised learning is between the supervised learning and the unsupervised learning methods where we have a large amount of input data, some of which are labeled and others are not. Many real-life learning problems are under this area of machine learning. This is because semi-supervised learning requires less human intervention since it utilizes very small amount of labeled data and a large amount of unlabeled data. Utilizing less labeled datasets is more appealing since such datasets are very difficult and expensive to gather and may require access to domain experts. Unlabeled datasets on the other hand are cheaper and easier to get access to (Swamynathan, 2017).

Both supervised and unsupervised learning techniques can be utilized to train the learning algorithm in semi-supervised learning. Unsupervised learning techniques can be used to unfold hidden structures and patterns in the input dataset. Whereas supervised learning techniques can be utilized to make guess predictions on the unlabeled data, feed the data back to the learning algorithm as training data and use gained knowledge to make predictions on new sets of data.

Knowledge Based Approach

It is a hybrid of acoustic phonetics and pattern recognition techniques. The artificial intelligence technique tries to automate the recognition process based on how a person uses his or her genius in perceiving and evaluating. Finally, a judgment is made based on the acoustic properties that have been measured. The artificial intelligence technique tries to automate the recognition process by mimicking how a person uses their intellect to visualize, analyze, and eventually decide on the observed acoustic data. In this method, the highly master system is often employed. This method was deemed un-workable; thus, an automated learning process was explored (Praphulla et al., 2015).

Deep Learning Approach

The deep learning approach makes use of neural networks which includes: Convolutional Neural Network, Recurrent Neural Network,Deep Neural Network for classification in Speech Emotion Recognition systems with spectrograms of two resolutions as input.Convolutional Neural Networks (CNN) are identical to regular neural networks, with the exception that they explicitly assume that the inputs are pictures, allowing us to embed particular qualities into the design. As a result, the forward function is easier to construct and the network's parameters are reduced. There are three dimensions to the neurons: width,height, and depth. The input layer, convolution layer, pooling layer, and fully connected layer are the four primary types of layers(Swamynathan, (2017)).

A feedback loop exists in a Recurrent Neural Network (RNN), which means that prior time steps are fed into the present phase. This sort of architecture creates sequences to mimic situations and generate synthetic data, making it an appropriate modeling solution for sequence data applications including spoken text mining, picture captioning, time series prediction, robot control, and language modeling (Swamynathan, 2017).

Deep Neural Network (DNN) is represented as a hierarchical (layered) organization of neurons (similar to the neurons in the brain) with connections to other neurons. These neurons pass a message or signal to other

neurons based on the received input and form a complex network that learns with some feedback mechanism(Swamynathan, (2017)).

2.3 Review Of Related Literature To Chatbots

This section of the projects gives reviews on different papers, projects, and journals published on works of designing and developing chatbot systems, and entails detailed processes, design patterns, and methodologies used. The review is meant to give more insight into what methods and approaches work best. These reviews also give further understanding of different domains and possibilities the problem topic can result in the general solutions for design and development.

2.3.1 Design And Implementation Of A Chatbot For E-Commerce

This study looks at how existing online stores, whether they sell goods or services, use conversational interaction. Because WooCommerce is the most popular e-Commerce solution technology, with 43 percent of the internet using it, we built the project around it. Furthermore, because Telegram is the most popular messenger in Iran, the bot is built using the Telegram API, but the same design can be implemented in a popup window of an internet browser or Facebook Messenger with a few changes. This paper presented the authors' design of a chatterbot for conversational commerce. The proposed design was created to improve user interaction in social media marketing and make social media marketing more effective by utilizing the quick order method; however, additional user research should be conducted to investigate the effectiveness of the proposed design. Furthermore, the implemented bot is limited to the WooCommerce shopping system; adding support for other shopping systems could increase the use of chat-commerce bots. Furthermore, the system is limited to the data source shops; connecting the system to other data sources could improve user satisfaction with the recommendation system(Asadi et al, (2018)).

2.3.2 Designing And Developing Bilingual Chatbot For Assisting Ethio-Telecom Customers On Customer Services

The paper was able to define a chatbot as an application that is used to converse with a computer using natural language. The purpose of this research is to create a bilingual chatbot that can assist Ethio telecom customers with customer service in both Amharic and Afaan Oromo. Ethio-telecom is an Ethiopian telecommunications company that is the country's largest provider of internet and telephone services. This paper also demonstrates a common issue that most users have with their telecommunications service provider.

In this study, the dataset was created in JSON format using both Amharic and Afaan Oromo languages by filtering frequently asked questions from the organization's website. The tf-learn DNN and Bi-LSTM models were used to create the model. We compared those models using accuracy metrics to select the best for the proposed system development. Based on this, the DNN model had an accuracy of 83.52 percent and the was Bi-LSTM model had an accuracy of 85.23 percent before applying the regularisation technique. The L2 regularisation technique was used to overcome the model overfitting challenge. Following regularisation, the model was evaluated using Accuracy, Precision, Recall, and F1-score yielding 82.6, 85.7, 82.6, and 87.7, respectively. In addition to metrics evaluation, the model was evaluated by humans. This assessment was used to determine two criteria: system performance and user acceptance. The model's performance was

assessed based on the number of queries it generates for the proposed system and the number of responses it generates. The proposed system's acceptability by users was the subject of the second evaluation. This evaluation was based on five criteria: attractiveness, response time, usability, efficiency, and system feasibility(Fekadu, (2021)).

2.3.3 Medbot: Conversational Artificial Intelligence-Powered Chatbot For Delivering Telehealth After Covid-19

This project demonstrated the use of chatbots to aid user communication and access through telecommunication for medical response known as Telemedicine. During the recent Coronavirus outbreak, medical practitioners can use telemedicine to connect with their patients while attempting to reduce COVID-19 transmission among patients and clinicians. During the pandemic, Telemedicine has the potential to assist by allowing patients to receive supportive care without physically visiting a hospital by using a conversational artificial intelligence-based application for their treatment. As a result, telehealth will rapidly and dramatically transform in-person care into remote patient consultation. Because of this, it developed a Multilingual Conversational Bot based on Natural Language Processing (NLP) to provide free primary healthcare education, information, advice to chronic. The study introduces a novel computer application that functions as a personal virtual doctor and has been specifically constructed and extensively trained to interact with patients in the same way that humans do. Based on a serverless architecture, this application aggregates a doctor's services by providing preventive measures, home remedies, interactive counseling sessions, healthcare tips, and symptoms for the most common diseases in rural India. The paper proposes a conversational bot called "Aapka Chikitsak" on Google Cloud Platform (GCP) for delivering telehealth in India, to increase patient's access to healthcare knowledge and leverage the potentials of artificial intelligence to bridge the demand and supply of human healthcare providers. As a result of this conversational application, the barriers to accessing healthcare facilities have been reduced(Bharti et al, (2020)).

2.3.4 An UMLs Based Chatbot For Medical Students

This work provides a chatbot system design, and the use of natural dialogue has a significant impact on the design of interactive tutoring systems for medical students. The nature of student queries can be constrained to a small set of task-domain-specific templates. This paper describes the creation of a chatbot for medical students using the open-source AIML-based Chatterbean. For generating query responses, we use the widely available Unified Medical Language System (UMLS) as the domain knowledge source. The AIML-powered chatbot is programmed to translate natural language queries into relevant SQL queries**.** The SQL queries are executed against the knowledge base, and the results are returned to the user via natural dialogue. The student survey was conducted to identify the various questions posed by students. The chatbot was created to answer common template questions. To generate responses for queries for which knowledge was not explicitly encoded, knowledge inference techniques were used. Three experts rated query responses on a 1-5 point Likert scale and agreed on a Pearson Correlation Coefficient of 0.54 and p 0.05. Experts gave the overall average rating of 3.4(Kazi et al, (2012)).

2.3.5 A System For Educational And Vocational Guidance In Morocco: Chatbot E-Orientation

Chatbots are currently being used in a variety of online applications, most notably for shopping and as personal assistants. This project demonstrates that chatbots can provide a variety of benefits, including personalization and instant availability 24 hours a day, seven days a week. These advantages of chatbots can be advantageous in the educational sector. They represent a new type of natural language human-machine interface. However, chatbots have received little attention in academia, for example, by providing organizational support for studies, courses, and exams. This field of study is still in its early stages in the scientific community; thus, in our article, we set up a chatbot in the field of educational and professional guidance based on John Holland's theory and the RIASEC questionnaire to determine the dominant type of personality of undergraduate and graduate students seeking to enter the job market(Zahour et al, (2020)).

2.3.6 Chatbots – An Interactive Technology For Personalized Communication, Transactions, And Services

This project viewed and demonstrated chatbots as a piece of new information, communication, and transaction channel that allows businesses to reach their target audience via messenger apps such as Facebook, WhatsApp, or WeChat. Chatbots, unlike traditional chats, are not handled by humans, but rather by software that guides users through conversations. The most recent chatbot developments in customer service and sales are impressive. However, little research on chatbots has been published in the field of public transportation thus far. Passengers can use chatbots to find timetables, purchase tickets, and have a personal, digital travel advisor provide real-time and context-relevant information about their trips. According to the project, chatbots in public transportation systems collect and provide various data about users and their journeys. Travel, product, service, content preferences, usage patterns, demographic, and location-based data are all examples. Chatbots have numerous benefits for businesses as well as mobile users. They enable new user touchpoints, increase convenience, decrease service, sales, and support costs, one-to-one marketing, new data collection, and deep learning. Smartphone users can contact a business at any time and from any location by using chatbots. The users of an investigated prototype who were interviewed are remarkably open to new mobile services and adapt quickly to this technology(Zumstein et al, (2017)).

2.3.7 An Interpretation And Implementation Of Chatbot Using Dialog Flow For An Education Domain

The project provides and demonstrates guidelines while also requiring the use of a simple chatbot system with a tool (DialogFlow). It defines a chatbot as a computer program that interacts with users through natural language or text, giving the user the impression that they are conversing with an assistant. To generate the required response, most chatbots use Artificial Intelligence (AI) techniques. Previously, chatbots gave the appearance of intelligence by using far simpler pattern matching. However, the authors attempted to implement a chatbot using Dialogflow technology in this paper, which will aid in the implementation of a bot for an education domain, specifically for the MCA Department. This bot will be useful for students who want to learn more about the MCA department(Hussain et al, (2018)).

2.3.8 Virtual Customer Service: Building Your Chatbot

This project demonstrates the creation of a chatbot system for customer relationship management. Nowadays, there is a significant evolution in the simulation of human-to-human conversations via chats. These chatbots increase organizational efficiency and improve the performance of customer and partner support. That is why they are becoming more popular on the internet. At the moment, there are several solutions for conversational virtual agents and chatbots. Many of these solutions include artificial intelligence, natural language processing, and comprehension systems built-in**.** The goal of this study is to create a road map for those who want to create a virtual assistant using a framework or a pre-built solution. This work compares several chatbot-related products, providing information that assists in determining the best solution for each case(Sousa et al, (2019)).

2.3.9 Design And Implementation Of A Chatbot In Online Higher Education Settings

This project considers Chatbots as a system that can be found in places where online users spend more time, such as text or mobile messengers and websites (Fichter & Wisniewski, 2017; Skerrett, 2017). One of its benefits is that communication with users is available 24 hours a day, seven days a week, and users can interact with bots at their leisure, usually anonymously. Furthermore, the process of answering questions is quick and simple, and links to various internal and external resources can be added *(*McNeal & Newyear, 2013). These features make chatbots more convenient and appealing to end-users. Skerrett (2017), on the other hand, contends that chatbot responses are limited by the variables included in their development and that, while many chatbots exist, not all of them are useful. If the chatbot fails, a human will contact the user at the end. Conversational chatbots can be found in a variety of industries, including entertainment, commerce, business, health, tourism, marketing, and education (Murad et al., 2017; Pradana, Goh, & Kumar, 2018). They contribute to these fields by increasing user satisfaction while assisting in less time. Chatbots, for example, answer frequently asked questions (FAQs) in business and marketing, providing unending customer service (Pradana, Goh, & Kumar, 2018). To avoid user annoyance, the bot should "identify when customers need to speak to live representatives as early in conversations as possible" (Arcand, 2017) and direct them to the appropriate person. In libraries, patrons inquire about book availability, location, operating hours, policies, and check-out procedures from librarians. The exchange experience with a chatbot is similar to that of a librarian; it can provide a self-service option for both face-to-face and online customers (McNeal et al, (2017)).

2.3.10 The Application Of Chatbot For Customer Service In E-Commerce

This work demonstrates the completion of a survey and research into the development of chatbots in an e-commerce environment. It describes how Rahman et al. (2017) proposed that chatbots, while still in their early stages of development, will be the next big thing as a tool in the era of conversational services as their use grows in the mainstream market. This study investigates the use of chatbots in customer service for an e-commerce website, based on our belief that customer service is critical to a company's growth. The project also shows the highest number of predictions regarding the use of chatbots is similar to the job desk that customer service has, which is to understand the problem that a customer may be experiencing and provide the best response that meets the customer's expectations (Parasuraman et al, 1991).

According to the survey that they conducted on April 1st, 2020, there were 102 respondents from people with the majority being students studying at Bina Nusantara University who filled out the form, with the result that while the use of chatbot is still considered uncommon for many of the respondents, the majority of people between the age group of 17 and older are not opposed to the idea of using a chatbot as a customer service. The project demonstrates that many youths are more inclined to support the use of technology in their daily lives because they believe it is more efficient and convenient to do so, and this case is to assist them in finding information(Wibowo et al, (2020)).

2.3.11 Development Of An E-Commerce Chatbot For A University Shopping Mall

As emphasized by this project, before this study, pattern-based chatbots had a dubious level of intelligence. Although Hebron is pattern-based, it has a fast learning rate, which means it picks up on any e-commerce item that is added to its database right away. It also responds quickly and effectively to new training phrases added to its database. The goal of this work was to design and implement a chatbot for Covenant University Shopping Mall. The chatbot's goal is to converse with students in a smart, accurate, and real-time manner. Students can chat with the bot about specific items they want to buy and pay for them online before going to the mall in this manner. The chatbot is accessible via portable mobile devices or computers, which students can access from anywhere and at any time on campus, resulting in 24-hour online service. (This research will alleviate the discomfort felt by Covenant University Community members when they travel to CUSM to source items only to discover that the desired items are either out of stock or unavailable. It is also suggested that more features be added to the chatbot, such as the delivery of paid items to student halls of residence; more training phrases be added to the chatbot to give the chatbot a better social outlook; and the addition of items to the chatbot database be automated, for example, by adding a barcode reader option to the chatbot. It would also be beneficial to test the chatbot on larger datasets(Oguntosin et al, (2021)).

2.3.12 A Chatbot System For Mental Healthcare Based On Sat Counseling Method

This project used the Mann–Whitney U test on their designed chatbot system to see if there were any differences in stress levels before the course between the two groups. There was no significant difference in any of the scale scores. There was a significant difference in self-esteem, STAI, and SDS score changes in the Chatbot course (self-esteem: $p = 0.024$, STAI: ($p = 0.038$), SDS: ($p = 0.043$). Regarding the questionnaire score based on the TAM model, the average scores for all scores in the Chatbot course were higher than the Web course result of the Mann–U Whitney's test on the scores showed that there was a significant difference in the score of PEOU ($p = 0.030$) and BI ($p = 0.027$) and a significant trend in the score of PU. In this study, the SAT method was converted to digital content; a self-guided mental healthcare course and a system utilizing a chatbot were developed, and a comparative evaluation experiment was conducted with a Web course lacking a chatbot as a control group. The results of the experiment suggest that using the chatbot boosts users' motivation and helps them reduce stress and that it is effective in the self-guided mental healthcare course. The study, however, shows that the sample of users in this study is small, and the user's motivation and stress reduction effect was evaluated based on a single use of the course. Future research should include an increase in the number of participants as well as a survey on the long-term use of the course(Kamita et al, (2019))..

2.3.13 A Customizable Multilingual Chatbot System For Customer Support

This paper proposed a customizable, multilingual chatbot system to assist organizations in self-developing their chatbot as customer service. A live chat and ticketing system is integrated into the chatbot service to allow agents to provide additional assistance to their customers. The system built in the project demonstrated that customers could queue for a live chat service, and agents could join the live chat from their respective agent panel or mobile application, demonstrating flexibility and ease of access to common services. For problems that cannot be resolved immediately, the ticketing system will be used, and the agent will be able to generate a tracking ID for the specific customer along with a problem description. Agents can respond to tickets via their panel, and customers will be notified via their email addresses(Kasinathan et al, (2021)).

2.3.14 AI-Based Chatbots In Customer Service And Their Effects On User Compliance

This project demonstrates how AI-based customer attendants have grown in popularity in a variety of settings and cost-saving opportunities. However, many users continue to have unsatisfactory interactions with chatbots (e.g., high failure rates), which may lead to skepticism and resistance to the technology, potentially preventing users from complying with recommendations and requests made by the chatbot. In this study, they conducted an online experiment to demonstrate that both verbal anthropomorphic design cues and the foot-in-the-door technique increase user compliance with a chatbot's request for service feedback. Our study is thus an initial step toward a better understanding of how AI-based customer attendants may improve user compliance by leveraging the effects of anthropomorphism and the need to remain consistent in the context of electronic markets and customer service. As a result, this study expands on prior knowledge of CAs as anthropomorphic information agents in customer service. They hope that our study will serve as a springboard for future research on CA compliance mechanisms and improving AI-based capabilities in and beyond electronic markets and customer self-service contexts(Adam et al, (2020)).

2.3.15 An Exploratory Study Of Customer Perceptions Of Usage Of Chatbots In The Hospitality Industry

This study looks at how customers perceive the use of chatbots in the hospitality industry. The study's focus is on hotels, particularly chain hotels, so that chatbots can be integrated with CRM solutions, allowing hotels and customers to gather information from a single source. The study also demonstrates how, in other sectors of the hospitality industry, such as restaurants, there is a lack of exposure to highly automated systems, causing customers to struggle to understand the concept of automated processes. One of the most significant challenges identified is that many players in the hospitality industry still do not understand artificial intelligence and how it can redefine the industry's future. According to the study, younger customers were more open to the use of technology and expected technology to be involved in at least one aspect of any service being provided, and they generally preferred conversing with virtual assistants such as iPhone Siri and Google Assistant over agents, unless complex solutions where required(Dash, (2019)).

2.3.16 Design And Implementation Of A Chatbot For E-Commerce

This paper presented the authors' design of a chatterbot for conversational commerce. The proposed design was created to improve user interaction in social media marketing and make social media marketing more effective by utilizing the quick order method; however, additional user research should be conducted to investigate the effectiveness of the proposed design. Furthermore, the implemented bot is limited to the WooCommerce shopping system; adding support for other shopping systems could increase the use of chat-commerce bots. Furthermore, the system is limited to the data source shops; connecting the system to other data sources could improve user satisfaction with the recommendation system(John, (2019)).

2.3.17 Designing Interactive Chatbot Development Systems

In this paper, they propose and evaluate design principles for guiding the design of interactive chatbot development systems to increase the engagement of domain experts. As a result, they carried out the first design cycle of our DSR project and proposed three DPs based on the interactivity effects model *(Sundar 2012; Sundar et al. 2015).* They then deduced six DFs and implemented the design in a chatbot development system. They demonstrated how to connect an existing chatbot to the system using their API and how to implement the proposed design in a web application that can be shared with respective domain experts. The evaluation of the web application revealed that the proposed DPs increase the perceived interactivity of the artifact and have a positive influence on the participants' subjective and objective engagement. They were also able to explain the increase in their engagement by demonstrating a mediation effect of perceived interactivity. This demonstrates that interactivity is a powerful motivator for users to interact with the system for a longer and more extensive period (Lalmas et al. 2014). As a result, their research demonstrates that existing descriptive knowledge in the form of the interactivity effects model is a strong foundation for informing the design of interactive artifacts intended to increase user engagement. As a result, the first design cycle of their DSR project contributes two things: prescriptive knowledge for designing interactive systems that increase engagement and a novel artifact in the form of an interactive chatbot development system(Asadi, (2018)).

2.3.18 A Tool Of Conversation: Chatbot

This study was able to describe a chatbot as one of the simplest ways to transport data from a computer without having to think of appropriate keywords to look up in a search or browse several web pages to collect information; users can easily type their query in natural language and retrieve information. Information about the design and implementation of the chatbot is presented in this paper**.** According to the results of the survey conducted in this paper, the development and improvement of chatbot design are occurring at an unpredictable rate due to the variety of methods and approaches used to design a chatbot. The project explains how chatbots are excellent tools for interacting with users quickly. It demonstrates how chatbots can help by entertaining users, saving time, and answering difficult-to-find questions. The project describes how chatbots should be simple and conversational, since there are many designs and approaches for creating a chatbot, it can be at odds with commercial considerations. Researchers need to interact and must agree on a common approach for designing a Chatbot. In this project, they investigated the evolution of chatbots as well as their applications in various fields. Furthermore, a comparison with other Chatbots in the project shows that a general-purpose Chatbot must be simple, user-friendly, easily

understood, and the knowledge base must be compact. Even though some commercial products have recently emerged, improvements must be made to find a common approach to designing a Chatbot(Dahiya, (2017)).

2.3.19 Role Of AI-Induced Chatbot In Enhancing Customer Relationship Management In The Banking Industry

This project's work exemplifies the efforts of other AI researchers to develop robot-mediated communication that mimics human conversational language and improves bot interpretation. With the advancement of machine learning algorithms, natural language processing (NLP), and speech recognition, a chatbot can be used on a variety of platforms. The project also described how Sony Bank provided an AI-induced chatbot in the business environment, delivering new value to customers. Customers may find it difficult to converse with robots, but the requested information will be obtained in less time. The selection of chatbots is critical because each scenario necessitates a different type of chatbot. The introduction of chatbots has improved the customer experience by improving existing services and smoothing the process of dealing with a large number of customers in providing information about offers and benefits of products such as house loans, card details, and so on. Then, existing customers have gotten an extra level of convenience from chatbots, such as product promotion with sample screening. Finally, the chatbots were streamlined to focus on handling customers' frequently asked questions, which needed to be automated due to customers repeatedly asking the same questions. The work done demonstrates that while the question is the same, the doubt is different for each customer. Each inquiry must be understood to guide customers, which necessitates the use of specific intelligence if a chatbot is used. The project also emphasized the use of the Finplex robot agent platform to respond to queries, which is based on an automatic reaction system developed by Fujitsu financial services. It also demonstrates that the banking industry is dependent on customers; therefore, it is preferable to incorporate cutting-edge technology in all operations, particularly in customer-related services. Along with internal bank competition, non-banking institutions have put pressure on the banking industry to stay current. A better customer experience will result in a stronger bond between the bank and its customers, which is a positive sign for business. According to recent studies, AI-powered chatbots have resulted in better customer conversations. As a result, financial institutions should take advantage of this latest advancement to improve customer service and loyalty(Satheesh, (2020)).

2.3.20 Bank Chat Bot – An Intelligent Assistant System Using Nlp And Machine Learning

The proposed system would be a stepping stone in having in place an intelligent query handling program that could in the next phases not just respond but self-learn to improve itself thereby increasing not just the quality of customer service but also reducing human load, increase in productivity and of course an increasing number of satisfied customers. This paper explains the dataset that they have prepared from FAQs of banks' websites, architecture, and methodology used for developing such chatbots. Also, this paper discusses the comparison of seven classification algorithms used for getting the class of input to the chatbot(Kulkarni et al, (2017)).

2.3.21 Implementation Of Chatbot In Online Commerce, And Open Innovation

This project describes the use of chatbots and what it brings in terms of modernity, efficiency, and proximity. The project then explains how chatbots can be used to optimize all company processes, saving resources and time, lowering costs, and improving quality. It provided an example of how a chatbot linked to Facebook can enable better segmentation of the target audience and thus simply capture a larger market. A chatbot on Facebook Messenger can generate more conversations and collect valuable information from customers, allowing company employees to intervene at the appropriate time. In addition to working 24 hours a day, 365 days a year, the company could reach out and expand its area of activity. Furthermore, it enables the filtering of potential customers based on their level of interest. Hence, not all prospective customers are equally likely to buy. The company can determine who to focus more or less effort on to increase sales, lowering the cost of acquiring leads. In terms of theoretical contribution based on the SOR model, the results obtained in this work extend previous research in the online environment, such as on the navigation structure via restrictive navigation bars. We provide features such as welcoming the customer, emoticons, and clarity of questions, which influence consumer behavior to elicit personal information; for example, customers provide their email through the chatbot and become potential leads for the company. This work presents various processes for developing an effective procedure for obtaining leads via a chatbot. The use of the ManyChat program for Facebook Messenger to support the study company's e-commerce is an important tool for carrying out online marketing tasks, which leads to significant cost savings and improves the experience for potential clients(Illescas-Manzano et am, (2021)).

2.3.22 Usage Of Chatbot As A New Digital Communication Tool For Customer Support: A Case Study On Banglalink

The project give a design of a system in the case of MITA, it functioned as an agent that interacted with users through natural language interaction. The knowledge base and pattern matching technique of Banglalink's web-enabled chatbot were examined in this case study in the project. The manual development of MITA's and other chatbots' knowledge is the main shortcoming. Several corpora were used to retrain MITA, revealing that a chatbot can be used for more than just entertainment. The project also emphasized how chatbots could be used to animate or visualize a corpus, to learn/practice English, Bangla, or other languages, or to access an information portal for answers to questions. This project shows how efficient to use chatbots to automate work processes(Alam et al, (2019)).

2.3.23 Evaluation Of Use Cases For Chatbots In CRM

The goal in this study was achieved and realized by gaining a better understanding of how bots can be used in the field of CRM. Bots have received a lot of attention recently. They are becoming more capable and provide intriguing CRM advantages. As a result, CRM users must understand whether and how bots can be used. The primary goal was to identify customer-facing bot use cases and value propositions. A secondary goal was to integrate such a bot with a CRM system. The first step toward achieving these objectives was to gain an understanding of the current state of bots and CRM. The majority of this project summarises the history of bots and CRM. Part of this included explaining the current buzz around bots and how they fit into CRM. The second step was

to identify and describe methodologies that would serve as a foundation and guidelines for conducting the research. It also briefly discusses the ethical considerations of this work. This paper describes how the research was carried out, with the main focus on use cases for bots in CRM and their respective value propositions. It gathered 148 use cases using a combination of systematic literature review, group discussions, and expert interviews. These use cases were analyzed, compared, and classified in a stepwise synthesis. The final classification provides an overview of many CRM chatbot use cases and their value proposition for the user. As a result, it answers the initial questions about CRM use cases and their value. It covers rather simple, well-known use case classes like 'Question-Answer' and those with obvious relevance for CRM like 'Sales Assistant'. It also contains unexplored but interesting use case classes like 'Onboarding' or 'Call People to Action'. A second classification summarises the features and actions that bots can perform in specific use cases. Both classifications are subject to expansion. A bot prototype was created during the second and minor phases of the research. The prototype's concept was based on some of the use case classes provided by the first part. This demonstrates how to use the use, case classes. Bot design and technology were also taken into account during the development process. Most importantly, the prototype used a REST API to integrate a Siebel CRM system in the back end. To provide a personalized experience, the bot reads customer data from the CRM system, such as when it greets the user by name or makes recommendations based on currently active contracts(Thenent et al, (2018)).

2.3.24 Design And Implementation Of A Chatbot In The Context Of Customer Support

This work, titled "The New Apps," tells the story of how chatbots were given a lot of responsibility. There was a significant trend with chatbots in 2016, a trend that has faded for the time being but is still lingering in the background. However, conversational agents have a long way to go before they can be used completely unsupervised. Nonetheless, as previously demonstrated, the specific domain of customer service is very prone to chatbot use and is already very systematic. Particularly in the field of Natural Language Processing, where new methods are being developed daily and older methods are becoming even better. The rise of chatbots has left an indelible mark on history and in consumers' minds. Furthermore, the roles of humans and bots may be reversed soon. To expand on that thought, we might see bots interacting with other bots using human language as a means of communication to improve humanity's quality of life one day. These subjects are already being investigated with mixed results. However, the project shows that the authors remain optimistic about these types of technology as they will shape the future of our societies. Perhaps in a few years, these technologies will be indistinguishable from humans. To ensure that these issues do not end up in the wrong hands, we must be cautious about how researchers approach them(Peters, (2018)).

2.3.25 Digital transformation in financial services provision: A Nigerian perspective to the adoption of chatbot

The contextualisation of this study in Nigeria is particularly worthwhile for several reasons. Firstly, Nigeria is the most populated country in Africa, with a population of over 200 million, accounting for about 16% of the total population of the continent. Secondly, the country is an economic power, as the World Bank (2019) reported that Nigeria is the second-largest economy in the continent after South Africa. Thirdly, despite being a huge economy, many people in the country are still

unbanked. And as such, it could be argued for a digital transformation that can bring financial inclusion, i.e. engaging the many unbanked into the banking system. According to The Global Findex Database, 60% of Nigerian adults have are unbanked as they do not have an account with a financial institution or a mobile money provider (Demirgüç-Kunt, et al., 2018). Nigeria and six other countries contribute nearly half of the globe's unbanked adult population of 1.7 billion, and 56% of that figure are women. Fourthly, since Nigeria is an emerging economy, which is facing institutional adversity and underlying and persistent economic challenges (Mogaji, 2020), the task of meeting the financial needs of this enormous population presents challenges. However, These challenges highlight a huge prospect of adopting chatbot as a digital transformation tool to aid consumer engagement with financial services; and ultimately this utilisation of chatbot can deliver significant business gains and increased access for consumers (Fitzgerald et al (2013)).

2.3.26 An Architectural Framework for Facebook Messenger Chatbot Enabled Home Appliance Control System

This article presents a framework for chatbot-controlled home appliance control system and was implemented by programming a Raspberry Pi using the Python language while the chatbot server was also implemented using a Node.js on JavaScript. The Raspberry Pi was connected to the chatbot server via Wi-Fi using a websockets protocol. The chatbot server is linked to Facebook Messenger using the Messenger Application Protocol Interface. Messages received at the chatbot server are analyzed with RasaNLU to classify the user's intention and extract necessary information which are sent over websocket to the connected Raspberry pi. The system was evaluated using control precision and percentage correct classification with both producing a significant level of acceptance. This work produced a Facebook Messenger chatbot-based framework capable of controlling Home Appliances remotely(Aina et al, (2019)).

2.3.27 An Approach Towards The Development Of A Hybrid Chatbot For Handling Students' Complaints

Chatbots are artificial intelligence programs that are able to act and maintain a certain level of conversation with real people. The study develops a chatbot system- CSC StudentBuddy (hosted using Facebook Messenger Platform with A Facebook Page acting as the bridge), to reduce these challenges in the Department of Computer Science, University of Benin, Benin City. Data was collected by the interview method. The respondents were 400 level/final year students and lecturers in the department. The chatbot was built with an easy-to- understand non-coding bot building platform- Chatfuel. The build-up of the chatbot was then conceptualized and birthed into a functional prototype. Then, usability validation testing and analysis were conducted with the respondents and feedback was received in form of surveys. The feedback generated was used to improve the bot(Chete, (2020)).

2.3.28 COVID-19 Chatbot Application Using The Natural Language Processing With Dialog-Flow

According to Adam et al, (2020) early research has demonstrated the benefits of using health care Chat-Bot, such as helping with diagnostic decision support promoting and increasing physical activity. In the occurrence of serious pandemics like novel Corona virus (COVID-19), health bots can be beneficial supplement to personal clinical care or immediate medications. The global health status has never been threatened such like the one posed by the outbreak of COVID-19 pandemic in this period of our civilization. Since the discovery of the Coronavirus (COVID-19), it has become a global pandemic, and a great challenge to hospitals or healthcare staff to manage the flow of the high number of cases (Jimoh et al, 2020). Quick access to quality and curated information is believed to help prevent/combat the soaring spread of COVID- 19, thus Chat-Bot are perfect for when users are looking for NCDC generic COVID-19 related information and screening. In line with the above, this research would seek design and implement a novel Artificial intelligent (natural language processing) Chabot agent application, using the Dialogflow framework that allows information access through text and voice interaction for users who are looking for NCDC generic COVID-19 related information and to help clarify users' status via symptoms provided(Shawulu et al, (2021)).

2.3.29 Development of Chatbot System for Disseminating Information in Higher Citadel of Learning.

The system proposed is designed to provide a medium for automating the manual process for getting information from the university helpdesk and making significant information available on the web-based system like information during and after admission process. In this paper, a chatbot system was designed, developed and deployed to the web server. The system is a web app developed with HTML, CSS, Python and AIML (Artificial Intelligence Mark-up Language). The developed chatbot system has been applied to automatically power the university's help desk in a user friendly and interactive manner. The web-based interface developed is simple for users or students to disseminate information about the university. Questions, answers, keywords, feedback messages and logs that are being stored on the database can be easily retrieved within a short period of time. The results showed that the developed chatbot system would help in no small way to reduce some of the challenges and difficulties facing access to information in our institutions(Adekunle et al, (2021)).

2.3.30 Why do Chatbots fail? A Critical Success Factors Analysis

Although chatbots are not a new technology (Schumaker et al, 2007) and the general availability of technology and tools is increasing significantly (Galitsky 2019), from the end-user side, there still is a high failure rate of developed chatbots that are unable to understand user input, do not react (Seeger and Heinzl 2021; Brandtzaeg and Følstad 2018, Følstad et al,2018; Filipczyk et al,2016) or cannot longer be found on their previous used communication channels. These negative and frustrating experiences in interacting with chatbots are still seen as one of the key challenges in the deployment of chatbots (van der Goot et al, 2021; Følstad et al, 2018). Despite this, there has been minimal qualitative research to date exploring the exact reasons why organizations are taking their chatbots offline permanently or no longer maintaining them. Chatbot failure is annoying not only from the user's point of view, but also from the chatbot provider's perspective, who has put a lot of work, time and money into the development, as well as from a global

perspective, as single negative chatbot experiences can damage the reputation of chatbots in general (van der Goot et al, 2021). Scientific literature considers various individual perspectives, such as single-case field reports from practitioners about challenges in deploying chatbots (Fiore et al,2019), characteristics relevant to failure or success based on user surveys (Rodríguez Cardona et al, 2021; Diederich et al, 2021; Mozafari et al,2021), and guidelines for improving chatbot elements (Brandtzaeg, Følstad (2018)). Seeger and Heinzl (2021) proved that high failure rates among customer service chatbots may harm customer trust and stimulate negative word of mouth. These authors recommend using 'anthropomorphic design elements' to avoid these effects (Seeger, Heinzl (2021)). An aggregation of all these technical, behavioral, and institutional aspects and perspectives on failure and success of chatbots is not yet available. A deeper understanding of the various factors from different stakeholders that impact chatbots' failures and success would increase the probability of a chatbot achieving success. For this reason, it is desirable to investigate the critical success factors (CSFs) affecting the development, deployment and usage of chatbots(Janssen et al, (2021)).

2.3.31 The role of perceived social presence in online shopping: The effects of chatbot appearance on perceived social presence, satisfaction and purchase intention

Chatbots have developed over time, and now they are the future of online customer service (Hervouët, 2017). In a survey from Aspect (2016), 49 per cent of 1000 consumers would prefer customer service interactions to be conducted via intelligent chatbots.So, in this study the effects of a human and animated appearance and an organizational logo will be examined. To the best of the researchers' knowledge, no other research has examined the effects of manipulating appearances (human or animated appearance or organizational logo) in human-chatbot interaction. Moreover, there is lack of knowledge on whether the use of different chatbot appearances influences user perceptions of chatbot interfaces differently. To discuss this gap, this research focuses on the theory of social presence, as this theory is identified as an essential factor in the design of chatbots in an online environment (Gnewuch, Adam, Morana,Maedche, 2018). Additionally, social presence refers to the degree to which users perceive one as being present via the mediated interface (Gunawardena, Zittle, 1997; Lim, Hwang, Kim, Biocca, 2015). Moreover, social presence occurs when users do not notice the para-authenticity of mediated humans and/or the artificiality of simulated non-human social actors (Lee, Jung, Kim, 2006). Previous research on social presence has shown that pictures can convey a personal presence in the same way as personal photographs can (Gefen et al, 2003).

2.3.32 Mobile based chatbot application for HIV/AIDS counseling using artificial intelligence markup language approach

Along with the development of technology and the desire of Spirit Paramacitta Foundation to provide information and counseling for the community related to HIV / AIDS using technology, this study aims to build the CHATBOT application for HIV / AIDS counseling. Chatbot or Chatterbots or chatter robots are software agents that simulate an entity that can interact in a conversation. Chatbot can be considered as a question-answer system that can provide knowledge provided by experts according to user requests [7]. Chatbot technology is not new thing, it's started when Joseph Weizenbaum's program ELIZA published in 1966 . Eliza used pattern matching and

substitution methodologies to simulate conversation. The key method of operation on ELIZA involves the recognition of cue words or phrases in the input and the output of corresponding pre-prepared or pre-programmed responses that can move the conversation forward in an apparently meaningful way.
This research used chatbot as a question-answer system that can provide information regarding HIV / AIDS to the general public or PLWHA. Chatbot answers like a counselor to questions from users. The knowledge base used in the CHATBOT application is built using the artificial intelligence markup language (AIML). Reason for choosing AIML because it is efficient and lightweight in use. Chatbot application is made based on the Android mobile operating system because it is widely used by Indonesians. Market share of the Android mobile operating system in Indonesia reaches 91.16% (Ardiana et al, (2020)).

2.3.33 Customer brand engagement through chatbots on bank websites–examining the antecedents and consequence

Currently many private and nationalized banks are deploying chatbots for connecting and communicating with customers. This technology is expected to dominate the banking sector in the future by improving customer service. However, the success of banking chatbots will be effective when customers are satisfied with the chatbots and engage in using them. To probe in to the question, this study investigates the antecedents and consequences of customer brand engagement in using banking chatbots, with the lens of diffusion of innovation theory. The antecedents include interactivity, time convenience, compatibility, complexity, observability, and trialability.
The paper presents theoretical and managerial implications which enable banks to strengthen customer engagement, satisfaction and brand usage intention through chatbots(Hari et al, (2021)).

2.3.34 A Multilingual African Embedding for FAQ Chatbots

In this paper, we introduce a chatbot for African institutions that is omnichannel, multilingual and multi-dialectal. This solution would solve the digi- tal communication with citizens using official gov- ernmental language and local dialects, avoiding fake news, understanding and replying using the relevant language or dialect to the user. Services would be digitized and hence, illiterate, and vulner- able citizens such as rural women and people that need social inclusion and resilience would have easier access to reliable information(Mabrouk et al, (2021)).

2.3.35 Emerging-market consumers' interactions with banking chatbots

The purpose of this study is to explore how consumers in emerging markets interact and engage with banking chatbots when conducting bank transactions. Based on qualitative research using semi-structured interviews with 36 Nigerian residents, this study demonstrates that the four factors of the unified theory of acceptance and use of technology (UTAUT) can explain how emerging-market consumers interact and engage with banking chatbots. While age and technological experience are significant components in facilitating the use of chatbots, perceived expertise, responsiveness and security were found to be particularly important to users. This study delves into user experiences with conversational interfaces, specifically focusing on emerging markets(Mogaji et al, (2021)).

2.3.36 Public service chatbots: Automating conversation with BBC News

Bots are one form of algorithmic technology that has found a place in the modern newsroom, with chatbots leading the way as news organisations seek to attract new audiences using conversational forms of journalism. Recent advances in artificial intelligence (AI) and machine learning (ML) have fuelled increasing experimentation with machine autonomy and there has been much hyperbole in the press about the extent and impact of this on journalism.(Jones, (2019)).

2.3.37 English Master AMMU: Chatbot for English Learning

English speaking skills can be improved in various ways. The most efficient way is to communicate with peers. It will help to improve speaking skills and reduce anxiety while talking. But most of the people do not feel great while talking to their friends as they feel a bit awkward. English Master AMMU (Automatic Mega-agent Managed User Guide) is a Google Assistant-based Chatbot that aims to improve users' spoken English skills. AMMU will interact with the user in a friendly way, as if they were speaking with a friend. Also, AMMU allows the user to study various basic and advanced portions of the English language, gives the user practice activities and quizzes. These combined features make AMMU different from other Google Assistant-based Chatbots(Gayathri, (2021)).

2.3.38 Text Messaging-Based Medical Diagnosis Using Natural Language Processing and Fuzzy Logic

The use of natural language processing (NLP) methods and their application to developing conversational systems for health diagnosis increases patients' access to medical knowledge. In this study, a chatbot service was developed for the Covenant University Doctor (CUDoctor) telehealth system based on fuzzy logic rules and fuzzy inference. The service focuses on assessing the symptoms of tropical diseases in Nigeria. Telegram Bot Application Programming Interface (API) was used to create the interconnection between the chatbot and the system, while Twilio API was used for interconnectivity between the system and a short messaging service (SMS) subscriber. The service uses the knowledge base consisting of known facts on diseases and symptoms acquired from medical ontologies. A fuzzy support vector machine (SVM) is used to effectively predict the disease based on the symptoms inputted. The inputs of the users are recognized by NLP and are forwarded to the CUDoctor for decision support. Finally, a notification message displaying the end of the diagnosis process is sent to the user. The result is a medical diagnosis system which provides a personalized diagnosis utilizing self-input from users to effectively diagnose diseases. The usability of the developed system was evaluated using the system usability scale (SUS), yielding a mean SUS score of 80.4, which indicates the overall positive evaluation(Damaševičius, (2020)).

2.3.39 The Relationship between Artificial Intelligence (AI) Quality, Customer Preference, Satisfaction and Continuous Usage Intention of e-Banking Services

This study was set to examine the relationship between artificial intelligence (AI) quality, customer preference, satisfaction and continuous usage intention of e-banking services powered by AI solution. To achieve this, a cross-sectional research design was employed to collect data from the bank customers using purposive snowball sampling. In total 274 responses were retrieved from the bank customers that agreed to participate in the study and 246 were useful for the final data analysis. Analysis was done using partial least square structural equation modeling (PLS-SEM) approach with aid of SmartPLS. A snowball non-probability sampling procedure was used for the data collection. The findings indicate that the AI quality have a positive effect on satisfaction and satisfaction has significant positive effect on the intention of continuing use of e-banking services. However, consumer preference was not significantly related to continuous usage intention of e-banking services. This is practically possible in Nigerian context. Because we observe that majority of the respondents does not have specific preference of using or interacting with e-banking services powered by AI as against those control by human interactions. The implication of this findings is that is the used of AI based services in the Nigerian context is still at it infancy level. Therefore, there is still much to learn from the experience of both planners, implementers and beneficiaries. This study is perhaps, the first of its kind to examine the relationships of AI chatbot quality on customer satisfaction and continuous usage intention of e-banking services. It is therefore, contributes to the development of literature in this direction(Dantsoho et al, (2021)).

2.3.40 Development of a Voice Chatbot for Payment Using Amazon Lex Service with Eyowo as the Payment Platform

This study aims at building a voice chatbot device that can be used for payment using Amazon Lex Service with Eyowo as the payment platform. The chatbot is built by leveraging on Artificial Intelligence (AI) technologies in the form of a service called Amazon Lex for configuring the bot with utterances and responses and Lambda Functions to validate the responses while carrying out the transactions by calling the Eyowo Application Programming Interface (API). A Raspberry pi single board computer is utilised as the medium of communication between the user and the chatbot. The Raspberry pi runs a script that collects voice input through a Universal Serial Bus (USB) microphone connected to it, which is sent to the Amazon Lex to be processed using Automatic Speech Recognition (ASR) and Natural Language Understanding (NLU). Then the chatbot sends back a suitable response to the user through the speaker connected to the Raspberry pi. The device brings comfort and security to the visually impaired by providing balance and cash transfers(Samuel et al, (2020)).

2.3.41 Chatbots impact on user experience

Cheng and Jiang (2020) examined how chatbots, driven by artificial intelligence, impact the user experience (Cheng & Jiang, 2020). It was discovered that satisfied chatbot users were positively affected both towards using chatbot services again, but also in terms of increased loyalty. However, if the customer perceives any potential risk in connection to chatbots, it can affect user satisfaction (Cheng & Jiang, 2020). Trivedi (2019) discusses the relationship between the customer's

experience of chatbots and customers feelings for the brand (Trivedi, 2019). Therefore, companies must consider the potential risks that new technology and data may entail for customers and how the perceived risk may affect the overall customer experience(Trivedi, (2019)).

2.3.42 Chatbots within customer service

According to Huang and Rust (2020), companies using chatbots within the field of customer service have increased (Huang, Rust, 2020). Morgan (2017) discusses how the development of artificial intelligence and chatbots could help companies to understand the connection between technology and customer satisfaction, and further how the technology could increase usability online. Usability and user experience, is fundamental for co-creation between customer experience and customer satisfaction (Morgan, 2017). Effective problem solving is one factor Vaslow (2018) argues for that could impact customer satisfaction. Furthermore, Vaslow (2018) discusses how effective ways of solving problems could impact customer satisfaction (Vaslow, 2018). Research made by Huang and Rust (2020) investigated how artificial intelligence is reshaping service and the tasks that previously have been related to human performed tasks (Huang ,Rust, 2020). Furthermore, a study made by Xu et al,(2020) investigated if customers perceived help from humans versus artificial intelligence differently (Xu et al, 2020). Artificial intelligence could be defined in several ways. However, Xu et al, (2020) have defined it from a customer service context as "a technology- enabled system for evaluating real-time service scenarios using data collected from digital and/or physical sources in order to provide personalised recommendations, alternatives, and solutions to customers' enquiries or problems, even very complex ones" (Xu et al, 2020). Moreover, it was discovered that customers' desire to use artificial intelligence for customer service varies depending on the complexity of the problem (Xu et al, 2020).
The research findings demonstrated that bank customers in Taiwan, to a higher degree, are willing to get service and help from artificial intelligence rather than from humans regarding low-complexity problems. Furthermore, Xu et al,(2020) argue for a connection between the problem's complexity and customer's willingness to use artificial intelligence (Xu et al, (2020)).

2.3.43 QUIZBOT: Implementation Of A Chatbot-Based Conversational Interface For Engaging Learning Contents

The advancements in chatbot technology presents an opportunity for a “personal learning assistant” that would significantly boost student’s academic performance. In this paper, we report the implementation of a conversational interface for engaging of learning contents using chatbot technology. Through a conversational interface, the prototype reminds the student of their study schedules, generate multiple choice questions from various study contents uploaded by the student, administer quizzes to help the students attain mastery of the contents(Ezenwoke et al, (2021)).

2.3.44 Evaluation of Modern Tools for an OMSCS Advisor Chatbot

The goal for this project is to reduce the response time from 24-48 hours to a few minutes using conversational chatbot technology.
A conversational chatbot is a tool that uses natural language processing (NLP) for human machine interaction (HMI). Historically, chatbots were used for entertainment. With the goal of tricking a

human into thinking they were talking to another human. Modern chatbots are tools used to convey information(Gregori, (2017)).

2.3.45 Use of recommendation engine and chatbot for depression consultancy platforms

As mentioned by Michele L McNeal David New year, chatbots have the trait of responding to different types of queries with the right information based on the aid of natural language processing (NLP). Based on test cases, chatbots are often programmed to be polite even when users use any rude or impolite language. Thus, in cases of depression, depressed people will never have to face a scenario whereby, chatbots become rude and less positive that could potential affect their mental health. In terms of medical technology, chatbots have been very vital. Some example is HealthTap and Your.md.
The algorithms that will be used to develop this chatbot will be importing libraries that acts as a depression therapist in order to keep the patients using the chatbot in a good mood. The dataset being used is obtained from Kaggle which consist of sets of tweets, classified into positive and negative, as the training data, in which 20% will be used as a cross validation data. The data will be preprocessed by using the Gensim library in which it removes hashtags, website links and user references in the tweets. A model is coded in python using Tenserflow, Keras and Gensim libraries. Then implemented a LSTM in the neural network model for identifying context in the tweet(Prasad et al, (2019)).

2.3.46 Chatbots as an approach for a faster enquiry handling process in the service industry

Nowadays, it is the main goal of organisations to create a customer service experience which customers are satisfied with and to create loyalty amongst them. To do so, chatbots can be used because they are available permanently, providing quick answers, they can promote offers and increase cross-and upselling (McCall, 2017), they can make individual recommendations to the customer, increase convenience, save overall costs and maximise customer engagement and outreach (Jain, 2016). Likewise, incorporating chatbots can bring along great potential to open up new opportunities concerning customer service. However, chatbots also carry along many limitations, especially because they have been in commercial use only for a few years(Weißensteiner, (2018)).

2.3.47 Survey On Chatbot Design Techniques In Speech Conversation Systems

Human-Computer Speech is gaining momentum as a technique of computer interaction. There has been a recent upsurge in speech based search engines and assistants such as Siri, Google Chrome and Cortana. Natural Language Processing (NLP) techniques such as NLTK for Python can be applied to analyse speech, and intelligent responses can be found by designing an engine to provide appropriate human like responses. This type of programme is called a Chatbot, which is the focus of this study. This paper presents a survey on the techniques used to design Chatbots and a comparison is made between different design techniques from nine carefully selected papers according to the main methods adopted. These papers are representative of the significant improvements in Chatbots in the last decade. The paper discusses the similarities and differences in the techniques and examines in particular the Loebner prize-winning Chatbots(Sameera A Abdul-Kader et al, (2015)) .

2.3.48 Towards A Chatbot For Digital Counselling

The aim of this paper is to outline the design of a chatbot to be used within mental health counselling. One of the main causes of the burden of disease worldwide is mental health problems. Mental health contributes to 28% of the total burden of disease, compared to 16% each for cancer and heart disease in the UK. Stress, anxiety or depression accounted for 15.8 million days of sickness absence across the UK in 2016. By 2018, the gap between the demand for mental health care and the resources the National Health Service (NHS) can provide is likely to widen, therefore providers are increasingly needing to find more cost-effective ways to deliver mental health care. Digital Interventions have been created to help with these issues, for example anxiety, stress and depression. Chatbots can be incorporated into digital interventions, or used as standalone interventions. Chatbots can be a more interactive experience for the user to receive information, or complete diagnostic tools, or to even be used for counselling. A demo chatbot was created using interactive emoji's and GIFs to improve the user experience when searching for online self-help tips. This chatbot will be further developed and incorporated into a full web based programme for mental health in the workplace. It is envisaged that the chatbot will be able to provide initial counselling, and lead users into the correct services or self-help information(Manik Sharma et al, (2019)).

2.3.49 A Deep Reinforcement Learning Chatbot

MILABOT: a deep reinforcement learning chatbot developed by the Montreal Institute for Learning Algorithms (MILA) for the Amazon Alexa Prize competition. MILABOT is capable of conversing with humans on popular small talk topics through both speech and text. The system consists of an ensemble of natural language generation and retrieval models, including template-based models, bag-of-words models, sequence-to-sequence neural network and latent variable neural network models. By applying reinforcement learning to crowdsourced data and real-world user interactions, the system has been trained to select an appropriate response from the models in its ensemble. The system has been evaluated through A/B testing with real-world users, where it performed significantly better than many competing systems. Due to its machine learning architecture, the system is likely to improve with additional data(Serban et al, (2017)).

2.3.50 Real conversations with artificial intelligence: A comparison between human–human online conversations and human–chatbot conversations

This paper analyzed how communication changes when people communicate with an intelligent agent as opposed to with another human. We compared 100 instant messaging conversations to 100 exchanges with the popular chatbot Cleverbot along seven dimensions: words per message, words per conversation, messages per conversation, word uniqueness, and use of profanity, shorthand, and emoticons. A MANOVA indicated that people communicated with the chatbot for longer durations (but with shorter messages) than they did with another human. Additionally, human–chatbot communication lacked much of the richness of vocabulary found in conversations among people, and exhibited greater profanity(Hill et al, (2015)).

2.4 Summary Of Research Review

The various publications and research works reviewed in this section concludes to gain knowledge on the development of chatbots. So many different chatbots were compared and found the most optimum way to create a chatbot. Chatbots in the healthcare, education, e-commerce domain e.t.c, were reviewed and useful information was gained from it. The pattern matching technique will help to create artifacts.

Conversational chatbots can be found in a variety of industries, including entertainment, commerce, business, health, tourism, marketing, and education (Murad et al, 2017; Pradana, Goh, Kumar, 2018). They contribute to these fields by increasing user satisfaction while assisting in less time. Chatbots, for example, answer frequently asked questions (FAQs) in business and marketing, providing unending customer service (Pradana, Goh, Kumar, 2018). To avoid user annoyance, the bot should "identify when customers need to speak to live representatives as early in conversations as possible" (Arcand, 2017) and direct them to the appropriate person.

Morgan (2017) discusses how the development of artificial intelligence and chatbots could help companies to understand the connection between technology and customer satisfaction, and further how the technology could increase usability online. Usability and user experience, is fundamental for co-creation between customer experience and customer satisfaction (Morgan, 2017).

Researchers need to interact and must agree on a common approach for designing a Chatbot. Several corpora were used to retrain MITA, revealing that a chatbot can be used for more than just entertainment.

conversational agents have a long way to go before they can be used completely unsupervised(Alam, Islam, Khan, (2019))

Rahman et al, (2017), proposed that chatbots, while still in their early stages of development, will be the next big thing as a tool in the era of conversational services as their use grows in the mainstream market.

A chatbot can be programmed without the aid of Machine Learning support and in this case, it is based on pure program logic. Chatbots have specific functions in terms of information retrieval (IR): there are virtual assistants, question-answer bots, and domain-specific bots. The goal of natural language processing (NLP) is to take the ASR's unstructured output and generate a structured representation of the text that includes spoken language understanding (SLU) or, in the case of text input, natural language understanding (NLU)(Dahiya, (2017)).

CHAPTER THREE

SYSTEM ANALYSIS AND METHODOLOGY

3.1 System Analysis

This chapter focuses on the analysis done as a stage in the development of a modern and conversational chatbot by using and improving previous solutions. The system is aimed at keeping the customers engaged while making a complaint or an inquiry for a solution to a problem that they may face on using their telecommunicator service. For the analysis of this work, the effort is to develop an analysis strategy by studying the currently used chatbot by MTN (Zigi Chatbot); the

effort is to gather the requirements for the development of the proposed system and then draft the system proposal at the end of the analysis. The system will be able to recommend solutions or give replies through context-based conversation with the customer and the design of this system will be achieved by using data analytics and data science methods to create NLP models, integrating the models by providing an interface which communication will take place between the user (customer) and the chatbot. Exploration of datasets through Data mining (gathering different real-life conversations had between a human customer and a human customer service representative) will take place thereby achieving a well-versed conversant chatbot in the field of MTN telecommunication service.

3.1.1 Outline Of Problem With The Existing **The problems associated with the existing system are;**

1. The interface used to provide the solutions is not well interactive to keep users engaged with the system(it is not user friendly).
2. The context of conversations which the chatbot can keep are not well versed and do give repeated or less conversational responses.
3. It is difficult to be understood by an average skilled computer user.
4. It doesn't truly interact with customers, understand their requirements based on the conversation, and offer relevant product recommendations.
5. The chatbot was trained with very limited dataset.

3.1.2 Description Of The Proposed System

The system this project aims and proposes is meant to facilitate some functionalities and features other existing system does not provide. The most focused part of the proposed system is aimed at practical functionality and usability for different users irrespective of their computer skills. The system will be developed using NLP and other machine learning algorithms to build the interactive chatbot, the user interface will be web-based to be available on most platform devices. Rather than an only or completely conversational bot, an interactive predefined solution to common inquiries.

3.3.1 Justification Of Proposed System

The justifications (advantages) of the proposed system are as follows:

1. There will be accessibility anywhere across the local country
2. The commonly shared problem encountered in telecommunication services people uses can be solved by providing one solution; making it more versatile to various types of problem.
3. It gives an interactive predefined solution to common inquiries.

3.1.3 Functional Requirement

The functional requirement describes or explains the behavior of the system; they define the functionalities within the software system. They also pick what the proposed system is expected to do or rather the services which are expected of the system to render to its users.

The functional requirements of the proposed system are as follows;

- Having an interface for user engagement and interaction.
- The system will be able to differentiate between problem-based and inquiry-based requests.
- Users should be able to access the system on their mobile smart devices.
- The system will produce a reliable path to solutions to users' problems and inquiries.

3.2 System Design

The system design explains the system operation and requirements. The reason for this phase, is to provide an understanding of how the system operates. It involves explaining the system's architecture, interfaces, input and output technique, modules, files, data, and database that satisfy the system's requirements.

3.2.1 Objectives of the Design

System design aims at achieving the following:
1. Physical and logical models of the system.
2. Interactive user interface (UI).
3. System's blueprint that satisfies all the requirements (both software and hardware).

3.2.2 Architectural Design Of An Intelligent Conversational Chatbot For Mtn Customer Service

The core component of the system is the NLP engine. The NLP engine works as a definitive component that converts the language structure that is passed into the system into actual intent and further processing. The NLP engine is divided into two subcomponents: the Intent Classifier and Entity Extractor. They convey the meaning of the input and take an action as an agent to fulfill the intent of the input, respectively. Derivations of meaning and intent behind inputs are computed with a function based on the knowledge base, with multiple inferences that are generated by the "model" trained. The knowledge base is a module that is in the form of questions and answers but has the characteristics of predefined intents.
A plugin component is attached to help communicate with the NLP engine and also relay message intent back to the input origin. The input origin is known as the "front end," while the plugin component is referred to as the "back end." The frontend is an interface for interaction with the common user. It provides familiar components to input messages and send them to the backend. The architectural design of the system is shown in the figure below (fig. 3.1).

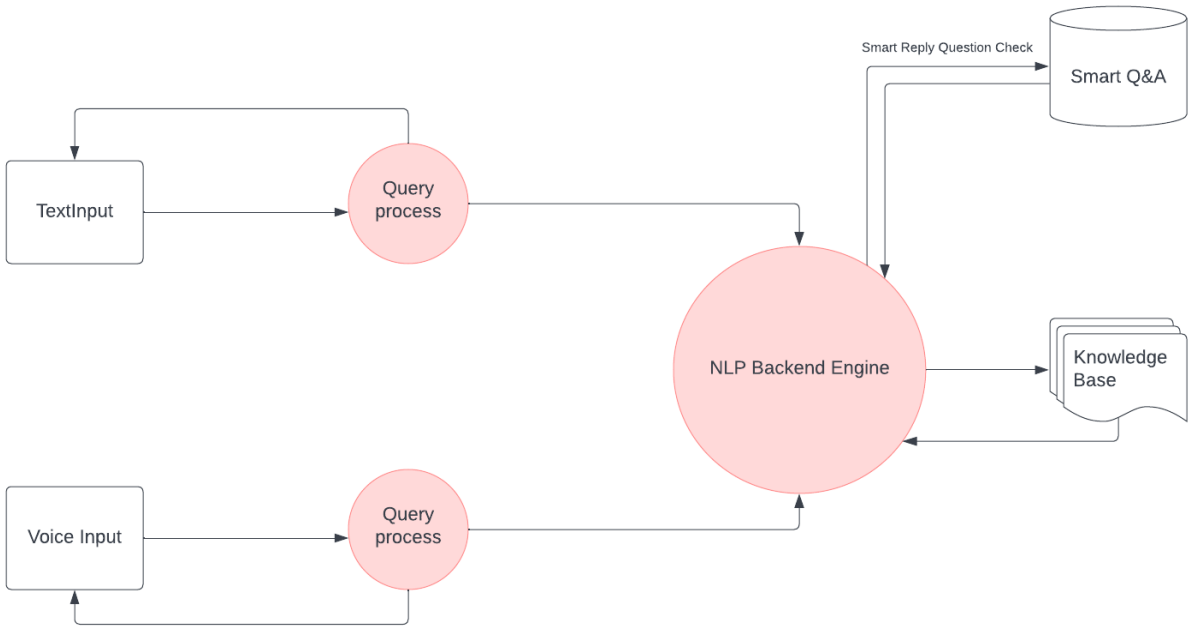

Fig 3.1 Architectural design of the system

3.2.3 Physical Design

The physical design of the system is related to the actual input and output processes of the system. It primarily focuses on how to enter data into the system, validate it, process it, and display the outcomes of the processes. It produces a workable system by defining design requirements that precisely outline the functions of the candidate system under consideration. It consists of three parts: user interface design, process design, and data design. As to its relation to this work, the user interface contains an input field to collect textual and voice data, send it over the network to the backend (a server that runs and communicates the NLP engine), and displays the generated response to the user..

3.2.3.1 Input Design

The input design has two control buttons, one for handling textual submission of messages and the other for instantiating an audio stream to send voice messages (which will be transcribed to text) and a text field that collects the text input.

Fig 3.2 Input design

3.2.3.2 Output Design

The output design of a system is essentially one of the most important aspects of any system as it is very necessary to know the representation of data that have been processed. When creating the output design of a system, the developers took into consideration the easily understandable ergonomics of devices and the user experience. Below is an output design for our project work. The input design of our system is shown in fig. 3.3. The right message bubble is the user's, and the left message bubble is the chatbot's response.

Fig 3.3 Output design

3.2.4 Logical Design

The system's logic is highly imperative and divided into a variety of abstraction models and modules, including authentication (which is not particularly sophisticated but is used to identify the current user who is chatting with the chatbot), messaging, memory state, and backend communication, which represent data flows, database design, and the inputs and outputs of the system), as well as messaging, memory state, and backend communication. Using Google Remote Procedural Call (gRPC) to communicate with the Google Cloud System.

```
lib > repository > base.dart
1  import 'package:dialogflow_grpc/dialogflow_grpc.dart';
2  import 'package:dialogflow_grpc/generated/google/cloud/dialogflow/v2beta1/session.pb.dart';
3  import 'package:flutter/services.dart';
4
5  import '../api/auth.dart';
6
7  class BaseRepository {
8    static BaseRepository? _instance;
9    static BaseRepository get instance => _instance ??= BaseRepository();
10   late DialogflowGrpcV2Beta1 _dialogflow;
11   Stream<StreamingDetectIntentResponse>? responseStream;
12   late InputConfigV2beta1 _dfAudioConfig;
13
14   Future<void> ensureInitialize() async {
15     final serviceAccount = ServiceAccount.fromString(
16         (await rootBundle.loadString('assets/credentials.json')));
17     _dialogflow = DialogflowGrpcV2Beta1.viaServiceAccount(serviceAccount);
18
19     //instantiate authentication
20     BotAuthtentication.initialize();
21
22     var biasList = SpeechContextV2Beta1(
23       phrases: [
24         'Dialogflow CX',
25         'Dialogflow Essentials',
26         'Action Builder',
27         'HIPAA'
28       ],
29       boost: 20.0,
30     );
31
32     // See: https://cloud.google.com/dialogflow/es/docs/reference/rpc/google.cloud.dialogflow.v2#google.cloud.dialogflow.v2.InputAudioConfig
33     _dfAudioConfig = InputConfigV2beta1(
34       encoding: 'AUDIO_ENCODING_LINEAR_16',
35       languageCode: 'en-US',
36       sampleRateHertz: 16000,
37       singleUtterance: false,
38       speechContexts: [biasList],
39     );
40   }
```

Fig 3.4 Logic design

Use Case Diagram

This diagram defines the system behavior based on the requirements that have been stated by the user. It goes further to also state a set of functionalities that the system should perform for a set of entities that it relates with. However, it does not describe the implementation of functionalities. Fig. 3.5 shows the use-case diagram of the proposed system.

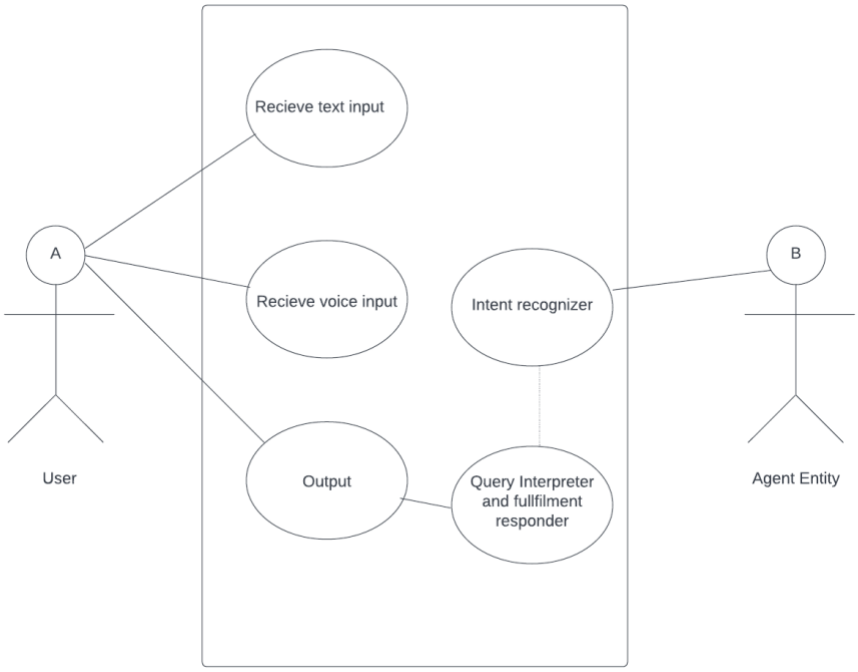

Fig 3.5 Use Case diagram of chatbot system

3.2.5 The User Interface Design

The user interface is designed for easy accessibility. to guide the user in all aspects of what the system does and how to use it quickly. The image below shows a minimal one-page onboarding screen that quickly shows what the system is all about.

Fig 3.6 user interface design

3.2.5.1 Main Menu

The main menu describes the interface of the home page in the system. It displays what the users will see and how to get into the system.

3.2.6 Database Design

The Database design for Chatbots is a set of processes that helps in the creation, implementation, and management of the database system. It creates the conceptual and physical designs of the database system.

3.2.6.1 Data Dictionary

During the development of the model, over 2000 historical data on frequently asked questions by customers from the MTN Company Head Office in Abuja and the internet to prepare our model. The major focus for this research is on a NLP(Natural Language Processing) aspect that deals with having definite responses to some questions with a guided area. The machine Learning algorithm that will be used for this research is Multinomial Naive Bayes, which is done by tokenizing sentences (that is, knowing count of words, or group of words and letters) to make a good answer/ response to questions.

3.2.7 Application Algorithm

Algorithms are the step-by-step means of solving a problem. Thus, relating it to this project, we can say our application algorithms are step-by-step means of moving through the sign language system so as to achieve the desired task, which in this case is to achieve a converted sign. In moving through our system, we have five steps.

Algorithm steps involve:

1. Retrieve the user input from the text input.
2. Create instance to connect with gRPC.
3. Send the processed data to Google Cloud using the instance.
4. Await for response (crequest made are asynchronous).
5. Display fulfillment response from Google Cloud.

3.2.7.1 Sequence Of Micro-Operations

The figure below shows the sequence of Micro-Operations and functional parts of the system works and therefore provides a functional chatbot system.

Fig 3.7 Sequence of Micro-Operations of the Chatbot System

3.3 Methodology

This section of the project describes the methodology adopted by following the system design to achieve the objectives of this project. The chosen methodology for this project is the CRISP-DM (Cross-industry Standard Process for Data Mining) .We employ the scientific methods identified with this methodology. The project is basically developed/built using Python programming language tools. NLP is used for the data analysis and utilizing deductions from the developed system.

3.3.1 Description Of The Selected Methodology

The selected methodology for the Development of an Intelligent Conversational Chatbot for MTN Customer Service is CRISP-DM(Cross-industry Standard Process for Data Mining) as aforementioned.

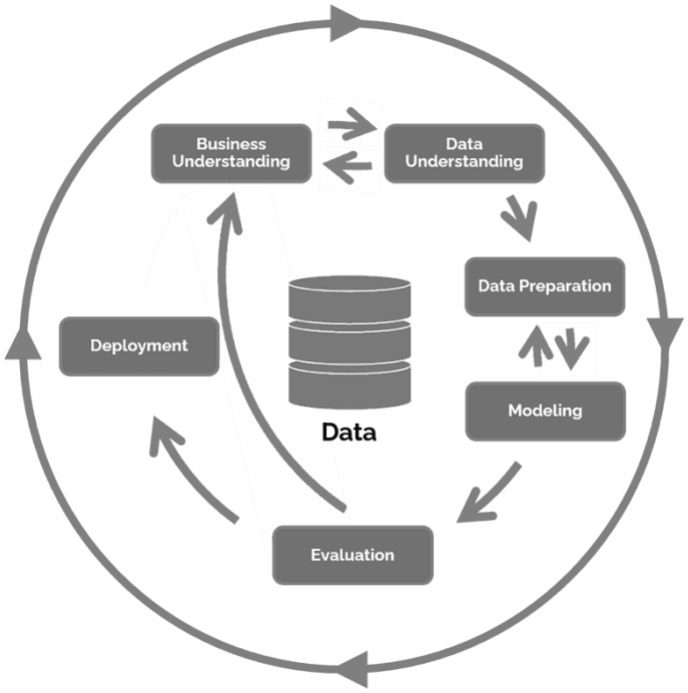

Fig 3.8 Cross-industry Standard Process for Data Mining(CRISP-DM)

CRISP-DM(Cross-industry Standard Process for Data Mining) is selected for the research due to its sequential approach to this problem and iteratively approach to the solution objectives.

The Cross Industry Standard Process for Data Mining (CRISP-DM) is a process model with six phases that naturally describes the data science life cycle. It's like a set of guardrails to help you plan, organize, and implement your data science (or machine learning) project.

The various stages of the methodology includes the following;

1. Business understanding – What does the business need?
2. Data understanding – What data do we have/need? Is it clean?
3. Data preparation – How do we organize the data for modeling?
4. Modeling – What modeling techniques should we apply?

5. Evaluation – Which model best meets the business objectives?
6. Deployment – How do stakeholders access the results?

3.3.2 Justification for the Selected Methodology

For this research project, we used the Cross-industry Standard Process for Data Mining(CRISP-DM) methodology for data mining. The methodology was selected because of the following reasons:

• Provides a guiding strategy for the development process

• Improves the reusability of software modules

• Easy integration of machine learning model on system for deployment.

CHAPTER FOUR
SYSTEM DESIGN AND IMPLEMENTATION

4.1 Systems Implementation

This design and implementation of this model aims to inform readers about the design and implementation of an intelligent conversational chatbot for MTN customer service as well as how it functions in relation to the system's operational requirements. In the first section, the architectural design of the proposed system is presented, along with a thorough analysis of the agents in an environment. The implementation of the system solution is discussed in the second section; the flowchart for the proposed system is discussed in the third section, and the implemented system is discussed in the fourth section.
The benefit of the system design is that it allocates the requirements (software and hardware) and enables the comprehension of the system that is to be implemented. It provides better structural management of the project from top to bottom. System designs enforce flexibility and modularity, which can help improve the current version of the system by creating a centralized model of a typical chatbot system.

4.1.1 Program Module Specification

The system developed in this research work however, is divided into two modules:
Training module and Testing module.
Training Module:
This module received training accented audio samples of new users and store them in db.py database to enable recognition during the test phase.
Testing Module:
This module validates the test sample by comparing its features with trained features and identifies it as the text that is closely match.

4.1.2 Implementation Tools Used

This project uses a number of tools to develop a system, which are:

1. Python, as a programming language.
2. Jupyter Notebook : An environment that uses cells (like block or component) structure to define the flow of the program.
3. NLK Library: A python library that has ML alrgorithms and can be used to train NLP models.
4. Pandas: Another python library which helps read and operate with the dataset.
5. VS code: A lightweight programming IDE which helps in writing and debugging codes.

4.1.3 Developmental Process

The statistical analysis of customer complaints to MTN and its customer service representatives was studied before the system was developed, which helped focus on the requirements and specifications the system needs to meet its goals. The chatbot system is then designed to fit a user who will require fewer steps to get solutions to the difficulties or issues they have with MTN Network as an internet and call provider. A

dataset of conversations in the form of questions and answers was mined to train the NLP engine. Then, in order to improve the system's usability, it was decided to make it available as a service via an Android mobile application. Before deployment can be done, a model was trained using the dataset that was mined and was tested to improve the accuracy of the model. The mobile app was built using the Flutter framework provided by Google, and the model was deployed on Google Cloud Server. The Google Cloud Server provides other features like an authentication system, which was used to supplement the application.

4.1.4 Physical Implementation/Deployment

• Open the application
• The user registers by entering registration details

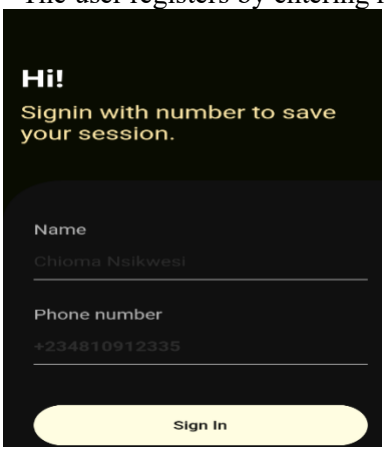

Fig 4.1: Sign In Form

• After registration, sign in with name and phone number acquired from registration process.
• After sign in, the user proceed to main page to chat with bot.

4.1.5 System Requirements

According to the Merriam-Webster dictionary, a requirement is something that is needed or wanted. In relation to this work, system requirements are the items or features that are needed to successfully make use of a system. The system requirements for a specific project are simply the specifications a device must have to use the software. Thus, for the sign language recognition system, we have the following software and hardware requirements.

4.1.5.1 Minimum Hardware Requirements

1. Smart Mobile Device
2. 16GB RAM and above
3. 320GB ROM and above.
4. Quad-core Intel Core i7
5. 8GB of GPU and above

4.1.5.2 Software Requirements

1. Latest Python execeutive interpreter
2. Programming IDE (VS Code)
3. Dart compiler and Flutter framework
4. Jupyter notebook extension.
5. Natural Language Kit.
6. Android Studio.

4.2 Presentation of Results

NLTK(The Natural Language Toolkit) with Python for the frontend was used to build the model.

4.3 Discussion of Results

For the model building we used NLTK(The Natural Language Toolkit) with Python for the frontend.The project is basically developed/built using Python programming language tools. NLP is used for the data analysis and utilizing deductions from the developed system.The back-end was developed using google cloud and gRPC(google remote procedural call) is the API used to link the front- end and the back- end while flutter is the framework that was used for the frontend.

The model was then deployed using Google Cloud.

A machine learning model that provides one solution primed to satisfy the modern customer, and that is a chatbot was successfully developed.

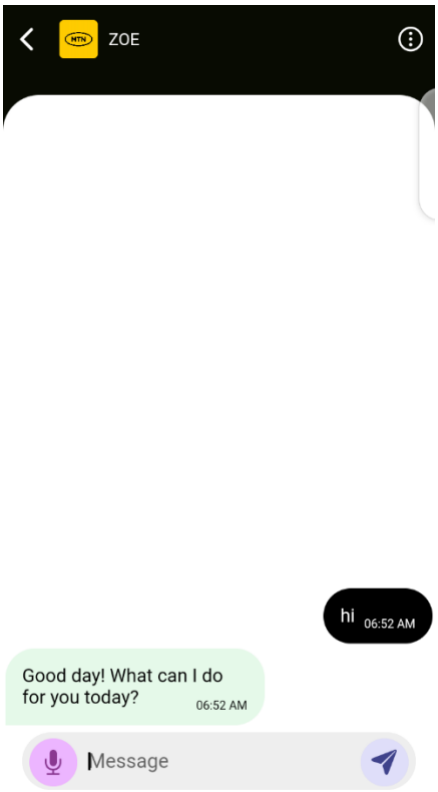

Fig 4.2 Chat Interface for MTN Customer Service

4.3.1 System Testing

This is a process of evaluating the software system opposing the user requirement elicitation and system specification. This ensures that the system error-free, produces the correct result, secured, reliable, efficient. The software system is tested at different levels or phases of testing; Unit testing, integration testing, system testing, and acceptance testing. The unit test was carried out in the course of the development process of the system to test each section of the program code. A system test was also carried out to test the requirement satisfaction of the system. Dynamic testing was also carried out to demonstrate the result of the running tests. Usability and accessibility testing were also carried out to show how the user can easily interact with the interface.

4.3.2 Changeover Procedure

Changeover procedures entails the change from one means of operation to another. The system does not serve as a replacement of the existing system components, rather it tends to improve the functionalities of the existing system.

4.4 Application or User Manual

The following are the steps required to enable users efficiently use the system:
Steps:
1. Open the web application.
2. Navigate to the Sign In section by clicking of the web application.
3. Click on the “Sign In” button to register as a new user or “Login” as an existing user.
4. Navigate to the main page to chat with bot.
5. Exit the application.

CHAPTER 5
SUMMARY, RECOMMENDATION, AND CONCLUSION

5.1 SUMMARY

A conversational chatbot is an effective way to provide many common solutions to many customers. When the system is used by the customer, they will experience easy access to solutions with just a few steps and without waiting on a line or spending much time to get simple actions done. In order to build a conversational chatbot for MTN customers, previous recordings of actual interactions between customers and real representatives were used to train NLP algorithms. In this research, NLK was used as an API for training and was able to create multiple instances of intent from the dataset. The system's model was deployed on Google Cloud Services, which also provides training through discussions with the users of the system. An interactive mobile application was built to serve the deployed model, which has easy-to-follow user interfaces to serve the solutions from the research.

5.1.1 Review Of Contributions And Achievements

As a result of conducting this research, I was able to

1. Obtain a dataset of over 2000 conversations with real customer representative and user of the network.
2. Prepare the dataset for training, validation and testing using NLTK library.
3. Create labels for the dataset
4. Train and validate the model on the dataset.
5. Test the model on a custom dataset.

5.2 RECOMMENDATIONS

The conversational chatbot system is highly recommended and can be extended to work for other telecommunication service providers. This research focuses on the user experience and user accessibility. Additional work can be done by making it more versatile in other languages over time, also the model can be re-trained with more dataset gathered over time.

5.2.1 Suggested Area of Future Research

Taking the achievements and limitations of the research into consideration, we intend to keep making improvements to the accuracy of the system by expanding the dataset. Future work should focus on:

- While there exists knowledge on current chatbot user groups, this needs to be updated as technology, services, and patterns of use evolve.
- Applying the tools used in this research to some other datasets and comparing the accuracy
- There is a need to move from studies of chatbot users in general to studies of chatbot users and behaviours for particular demographics, domains, or contexts

5.3 CONCLUSION

At the end of this work, we successfully developed a machine learning model that provides one solution primed to satisfy the modern customer, and that is a chatbot. With this chatbot, MTN organization can easily offer high-quality support and conflict resolution at any time of the day, and for a large quantity of customers simultaneously.This model was
trained with over 2000 dataset.

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